THE HIGHER DERIVED FUNCTORS OF THE ADEQUATOR VANISH ON ADEQUATE MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The description of modules on the affine site via presheaves, and the agreement of cohomology for adequate modules. Cohomology on Sites, Lemma sites-cohomology-lemma-include-modules and Same chapter, Lemma lemma-same-cohomology-adequate, invoked at proof lines 2547-2551. Show that j of an injective resolution is exact.
- An injective module is an injective presheaf of modules. Cohomology on Sites, Lemma sites-cohomology-lemma-injective-module-injective-presheaf, invoked at proof lines 2552-2555. Lets the resolution serve as an acyclic resolution for Q.
- The vanishing of the higher derived functors of Q. Same chapter, Lemma lemma-RQ-zero, invoked at proof lines 2556-2558. Gives the result.

Source reference. Original lemma 06ZI: sources/stacks-adequate/pinned-original-source.tex, source lines 1591–1591. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06ZS: sources/stacks-adequate/pinned-original-source.tex, source lines 2474–2474. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06VM: sources/stacks-adequate/pinned-original-source.tex, source lines 1896–1896. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06YK: sources/stacks-chow/tags.txt, source lines 8889–8889. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03FB: sources/stacks-chow/tags.txt, source lines 4486–4486. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $U = \mathrm{Spec}(A)$ be an affine scheme. For any object $\mathcal{F}$ of $Adeq(\mathcal{O})$ we have $R^pA(\mathcal{F}) = 0$ for all $p > 0$ where A is as in Lemma 06ZS.*

Proof. With notation as in the proof of Lemma 06ZS choose an injective resolution $k(\mathcal{F}) \to \mathcal{I}^\bullet$ in the category of $\mathcal{O}$-modules on $(Aff/U)_\tau$. By Cohomology on Sites, Lemmas 06YK and Lemma 06VM the complex $j(\mathcal{I}^\bullet)$ is exact. On the other hand, each $j(\mathcal{I}^n)$ is an injective object of the category of presheaves of modules by Cohomology on Sites, Lemma 03FB. It follows that $R^pA(\mathcal{F}) = R^pQ(j(k(\mathcal{F})))$. Hence the result now follows from Lemma 06ZI. □